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### Micro gas generator and method for producing a micro gas generator

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#### Abstract

The invention describes a micro gas generator (10) for a vehicle safety system comprising a plastic housing part (20) and a metallic insert (22), wherein the plastic material forming the housing part (20) is injection-molded around the insert (22) and, thus, the latter is positively connected to the housing part (20), and wherein a seat (24) in which an igniter (14) is inserted is formed in the housing part (20). In addition, a method of manufacturing a micro gas generator (10) is described.

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## Background/Summary

## RELATED APPLICATIONS

(1) This application is a U.S. National Stage Application filed under 35 USC 371 claiming priority to Serial No. PCT/EP2022/063844, filed on 23 May 2022; which claims priority from German Patent Application DE 10 2021 113 570.2, filed 26 May 2021, the entireties of which are hereby incorporated herein by reference.

## TECHNICAL FIELD

(2) The invention relates to a micro gas generator for a vehicle safety system and a method of manufacturing a micro gas generator of this type.

## BACKGROUND

(3) In vehicle safety systems, micro gas generators serve for triggering quick events, for example in belt tensioners, airbags, hood props etc.

(4) Due to the necessary short reaction times of few milliseconds, micro gas generators are usually operated pyrotechnically. Accordingly, gases formed during quick combustion are used to reach a very fast movement, e.g., of a piston or a fabric, or the break-up of a membrane by the resulting pressure.

(5) For triggering a pyrotechnical unit, an igniter which is usually ignited with a defined electric pulse is required. The pressure generated by an igniter is frequently boosted by means of propelling charges.

(6) Consequently, the igniter must be accommodated in a holder which resists the prevailing pressures.

(7) There are known machined metallic holders which are very cost-intensive, however, due to the involved plug interface. In addition, it is known to injection-mold plastic material around the igniter when manufacturing the micro gas generator, the plastic material simultaneously forming a plug interface for connection to the vehicle wire harness and an igniter support. It is a drawback here that the handling of pyrotechnical igniters requires complex protective measures, which has an aggravating effect on the process of injection-molding around the igniter. In particular technically more sophisticated injection-molding processes with a plug interface by the manufacturer require, apart from an expertise in fabrication methods, further knowledge for the handling and the processing of pyrotechnical elements.

## SUMMARY

(8) Therefore, it is an object of the present invention to facilitate the manufacture of a micro gas generator.

(9) According to the invention, this object is achieved by a micro gas generator for a vehicle safety system comprising a plastic housing part and a metallic insert, wherein the plastic material forming the housing part is injection-molded around the insert and, thus, the insert is positively connected to the housing part, and wherein a seat in which an igniter is inserted is configured in the housing part.

(10) A micro gas generator of this type offers the advantage that the fabrication of the plastic housing part and the fabrication of the igniter can be carried out completely separately from each other. In this way, the manufacture of the housing part is considerably facilitated as during injection-molding of the plastic housing part no protective measures due to the igniter have to be taken. This has a beneficial effect on the manufacturing costs of the housing part and, thus, on the total manufacturing costs of the micro gas generator.

(11) The plastic housing part along with the metallic insert forms a support of the micro gas generator, the metallic insert improving the mechanical stability of the support. In addition, the metallic insert can serve for fastening the micro gas generator.

(12) The metallic insert preferably cannot be detached non-destructively from the plastic housing part.

(13) According to the invention, the igniter is inserted into the support, specifically into the plastic

housing part, only after the support is completed.

(14) Preferably, the seat is configured so that, after being inserted, the igniter is held frictionally in the seat.

(15) According to one embodiment, plastic material can also be injection-molded partially around the igniter. More precisely, a portion of an ignition head of the igniter as well as an area of the electrical contacts of the igniter adjacent to the ignition head can be injection-molded. In this way, the electrical contacts are stabilized which is of advantage during handling and during assembly of the igniter. In particular, the stabilization of the electrical contacts reduces the risk of the contacts being bent.

(16) The seat for the ignition is preferably formed at least partly by a hollow-cylindrical portion of the housing part. Such a seat can be manufactured in a particularly simple manner.

(17) According to one embodiment, a fixing cap is attached to the hollow-cylindrical portion in which the igniter is received, or a fixing cap is inserted into the hollow-cylindrical portion. The fixing cap retains the igniter tightly on the housing part and prevents the igniter from detaching from the housing part when the micro gas generator is triggered.

(18) In order to obtain a sufficiently stable connection between the fixing cap and the housing part, the fixing cap is locked, glued and/or welded to the housing part.

(19) At a bottom of the fixing cap there is formed, for example, an elevation, specifically an annular elevation, which biases the igniter against the housing part. In this way, better sealing of the micro gas generator can be achieved.

(20) By means of the elevation, the fixing cap presses preferably only onto rigid areas on the igniter so that the igniter is prevented from being damaged. In particular, the fixing cap presses onto an edge area of the igniter by means of the elevation.

(21) If plastic material is injection-molded around the igniter, the igniter can be connected to the housing part by welding, locking and/or gluing the plastic material injection-molded to the igniter to the housing part. The fixing cap is not required in this case.

(22) According to one embodiment, a propelling charge container is attached to the housing part, the propelling charge container being configured such that the igniter is fixed in the seat by the propelling charge container. The use of a propelling charge container offers the advantage that the constant distance between the igniter and the propelling charge and, thus, the pressure generated by the igniter can be reproducibly increased by igniting the propelling charge. As the igniter is fixed in the seat by the propelling charge container, no separate fixing cap is required to fix the igniter.

(23) For example, the propelling charge container is held on the housing part by means of a locking connection. A locking connection enables particularly quick and simple assembly of the propelling charge container.

(24) As an alternative or in addition, the propelling charge container may be integrally connected to the housing part and specifically welded to the housing part. An integral connection helps obtain an excellent sealing and a particularly tight connection between the housing part and the propelling charge container.

(25) The igniter includes specifically an ignition head, wherein at an end face of the ignition head a collar projecting in the axial direction toward the housing part is formed and an annular seat for the collar is provided for in the housing part. The annular seat is formed as a negative form of the collar, for example. This offers the advantage that, when the igniter is inserted, a labyrinth effect occurs, causing the sealing of the micro gas generator to be improved.

(26) The collar preferably extends along an outer edge of the ignition head.

(27) According to one embodiment, the seat for the collar of the ignition head is partly formed by an annular projection extending in the axial direction toward the igniter base which is adjacent specifically to an inner face of the collar of the ignition head. When the ignition head is inserted into the seat and is pressed against the projection in the axial direction, a sealing surface is formed at the contact point. The sealing is additionally improved by the annular projection abutting on an

inner face of the collar of the ignition head.

(28) The projection can taper toward the ignition head, thereby the flexibility of the projection being improved.

(29) According to one embodiment, a stop is provided which delimits an insertion depth of the igniter. In this way, the igniter can be positioned particularly easily in a defined position in the housing part.

(30) Optionally, a sealing, and specifically an O-ring, is interposed between the igniter and the housing part. This improves additionally the sealing of the micro gas generator.

(31) The metallic insert is a disk projecting radially from the housing part, for example. The metallic insert can be connected and/or enclosed by an integral connection to a corresponding counter-surface of a vehicle safety structure, can be fastened to or bear against said structure.

(32) The metallic insert particularly is a punched metal disk.

(33) The object is further achieved, according to the invention, by a method for manufacturing a micro gas generator, wherein a metallic insert is placed into a tool mold and a plastic housing part is manufactured in the tool mold by injection-molding. The metallic insert is overmolded so that it is positively connected to the housing part. The plastic housing part includes a seat for an igniter, wherein an igniter is inserted into the seat after injection-molding.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) Further advantages and features of the invention will be evident from the following description and from the attached drawings which are referred to, and wherein:

(2) FIG. 1 shows a support and an igniter of a micro gas generator according to the invention,

(3) FIG. 2 shows parts of a micro gas generator according to the invention,

(4) FIG. 3 shows the partial assembly of a micro gas generator according to the invention,

(5) FIG. 4 shows a detail view in the area of the fixing cap,

(6) FIG. 5 shows a micro gas generator according to the invention as set forth in a further embodiment, and

(7) FIG. 6 shows another micro gas generator according to the invention,

(8) FIG. 7 shows yet another micro gas generator according to the invention, and

(9) FIG. 8 shows the igniter of the micro gas generator of FIG. 7.

### DESCRIPTION

(10) FIG. 1 illustrates parts of a micro gas generator **10**. In particular, a support **12** and an igniter **14** of the micro gas generator **10** are shown, the support **12** and the igniter **14** being provided separately from each other.

(11) The igniter **14** includes an ignition head **16** and electrical contacts in the form of pins **18**.

(12) A collar **19** extending in the axial direction is integrally formed on an end face of the ignition head **16**.

(13) The support **12** is formed by a plastic housing part **20** and a metallic insert **22**, wherein the plastic material forming the housing part **20** is injection-molded around the insert **22** and, thus, the latter is positively connected to the housing part **20**. In particular, the metallic insert **22** is overmolded in such a way that it cannot be detached non-destructively from the housing part **20**.

(14) The metallic insert **22** is a disk projecting radially from the housing part **20**.

(15) In the housing part **20**, a seat **24** is formed into which the igniter **14** can be inserted. On the pin side, a shaping of the plastic material is suitable for locking a plug for connection to the vehicle wire harness.

(16) The seat **24** for the igniter **14** is formed at least partly by a hollow-cylindrical portion **26**. In the hollow-cylindrical portion **26** the ignition head **16** can be received.

(17) Moreover, the seat **24** includes two recesses **27** in which the pins **18** of the igniter **14** can be received.

(18) The pins **18** extend, in the inserted state of the igniter **14** (see FIG. 2), through the support **12**, more precisely through the recesses **27**, and are accessible from one side of the support **12** for electrical contacting of the igniter **14**.

(19) In addition, an annular seat **28** for the collar **19** of the igniter **14** is formed in the housing part **20**.

(20) The annular seat **28** in the illustrated embodiment is formed by an annular projection **30** being provided which extends coaxially with a wall of the hollow-cylindrical portion **26** within the portion **26**.

(21) The annular projection **30** is tapered toward its free end.

(22) Accordingly, the taper **32** can extend over the whole length of the projection **30** or, as shown in the Figures, merely over a portion of the projection **30** which is adjacent to the free end of the projection **30**. As the taper **32** extends only over a part of the projection **30**, the projection **30** has a stabler design.

(23) The annular projection **30** at the same time forms a stop **34** delimiting an insertion depth of the igniter **14**. More precisely, the stop **34** is formed by the free end of the projection **30**.

(24) The stop **34** can also be formed at a different position of the seat **24**, however, such as by the bottom of the annular seat **28**. In the illustrated embodiment, there is provided an axial distance between the bottom of the annular seat **28** and the collar **19** of the igniter **14**, however, so that a gasket (see FIG. 3) can be inserted between the igniter **14** and the housing part **20**.

(25) FIG. 2 illustrates the parts of the micro gas generator **10** of FIG. 1, the igniter **14** being mounted on the support **12**, specifically on the housing part **20**.

(26) The igniter **14** is held frictionally in the seat **24**. In particular, the ignition head **16** is in contact with an inner wall of the hollow-cylindrical portion **26**, thereby obtaining, in addition to a friction force, also a sealing of the micro gas generator **10**.

(27) When the igniter **14** is inserted in the housing part **20**, as shown in FIG. 2, the ignition head **16** abuts against the stop **34**.

(28) The collar **19** of the igniter **14** is provided in the annular seat **28**, wherein the projection **30** may abut on an inner face of the collar **19**. In this way, a labyrinth effect occurs which ensures improved sealing of the micro gas generator **10**.

(29) In FIG. 2, additionally a fixing cap **40** is illustrated which can be arranged on the hollow-cylindrical portion **26**, more precisely can be attached to the latter so as to hold the igniter **14** tightly on the housing part **20**.

(30) For this purpose, connectors may be provided, such as locking elements by means of which the fixing cap **40** can be retained on the housing part **20**. As an alternative, a bayonet lock or a thread may be provided. As another alternative, on an inner face of the fixing cap **40** teeth may be formed, specifically a so-called fir-tree structure, the teeth being crimped when the fixing cap **40** is attached to the hollow-cylindrical portion **26**. In this way, a holding and sealing function is achieved. Such a fir-tree structure is schematically shown in FIG. 4 which illustrates a detail view in the area of a contact surface of the fixing cap **40** with the hollow-cylindrical portion **26** of the housing part **20**. The other connectors are not shown in the Figures for convenience.

(31) As an alternative or in addition, the fixing cap **40** can be welded to the housing part **20**. The welding can be performed by laser welding, vibration welding etc., for example.

(32) An elevation **42** is formed at a bottom of the fixing cap **40**.

(33) The elevation **42** is annular when viewed in a top view onto the bottom of the fixing cap **40**.

(34) The elevation **42** serves for biasing the igniter **14** against the housing part **20** when the fixing cap **40** is attached to the housing part **20**. This, too, serves for better sealing of the micro gas generator **10**.

(35) Inside the elevation **42**, the fixing cap **40** has an area of a small wall thickness. The wall

thickness and the geometry are selected particularly so that the fixing cap **40** can tear in a defined manner when the micro gas generator **10** is triggered so that gas can escape from the micro gas generator **10**.

(36) FIG. **3** illustrates a micro gas generator **10** in an assembled state with the fixing cap **40** being attached.

(37) In addition, in FIG. **3** a gasket **44** in the form of an O-ring is shown. The gasket **44** is disposed in the annular seat **28** and is compressed by the collar **19** of the igniter **14** when the latter is disposed in the seat **24**.

(38) As is schematically illustrated in FIG. **5**, a propelling charge container **46** can be provided. A propelling charge can increase the pressure formed by the ignition.

(39) The propelling charge container **46** can simultaneously serve for fixing the igniter **14**. For this purpose, the propelling charge container **46** can have a radially inwardly protruding structure **48** which biases the igniter **14** against the housing part **20**.

(40) The propelling charge container **46** can be connected, similar to the fixing cap **40**, to the housing part **20** by means of a locking connection, a bayonet lock, welding or in a different way.

(41) In an alternative embodiment, a propelling charge can be arranged on the ignition head **16**.

(42) FIG. **6** shows another embodiment of a micro gas generator **10**.

(43) The micro gas generator **10** shown in FIG. **6** differs from the micro gas generator **10** illustrated in FIGS. **1** to **3** in that the fixing cap **40** is not attached to, but is inserted into the hollow-cylindrical portion **26**. That is, the fixing cap **40** is attached to the ignition head **16**.

(44) FIG. **7** illustrates another embodiment of a micro gas generator **10**.

(45) The micro gas generator **10** illustrated in FIG. **7** differs from the above-described micro gas generators **10** substantially by the shape of the igniter **14** and a seat **24** in the housing part **20** which is appropriately adapted to the shape of the igniter **14**.

(46) The igniter **14** of the micro gas generator **10** is shown in FIG. **8**.

(47) The igniter **14** is provided with a plastic overmolding **50**. In particular, plastic material is injection-molded around a portion of the ignition head **16** as well as around the pins **18** in a portion adjacent to the ignition head **16**.

(48) The plastic overmolding **50**, on the one hand, serves to stabilize the pins **18** so as to prevent the pins **18** from being bent during handling or assembly.

(49) Furthermore, the plastic overmolding **50**, particularly the portion of the plastic overmolding **50** disposed on the ignition head **16**, serves for fastening the igniter **14** to the housing part **20**.

(50) For example, the plastic overmolding **50** is integrally connected to the housing part **20**. This can be done, as already above-described in conjunction with the fastening of the fixing cap **40** to the housing part **20**, by welding, specifically by laser welding, vibration welding etc.

(51) An integral connection of the plastic overmolding **50** to the housing part **20** is preferably made along the hollow-cylindrical portion **26**.

(52) As an alternative or in addition, locking elements which engage in a corresponding seat on the housing part **20** can be provided on the plastic overmolding **50**. For convenience, they are not illustrated in FIGS. **7** and **8**.

## Claims

1. A micro gas generator (**10**) for a vehicle safety system comprising a plastic housing part (**20**) and a metallic insert (**22**), wherein the plastic material forming the housing part (**20**) is injection-molded around the insert (**22**) so that the insert is positively connected to the housing part (**20**), and wherein a seat (**24**) in which an igniter (**14**) is inserted is formed in the housing part (**20**), the igniter (**14**) having an ignition head (**16**), wherein a collar (**19**) projecting in the axial direction toward the housing part (**20**) is formed on an end face of the ignition head (**16**) and an annular seat (**28**) for the collar (**19**) is formed in the housing part (**20**), the seat (**28**) for the collar (**19**) of the

ignition head (16) being partly formed by an annular projection (30) extending in the axial direction toward the ignition head (16) which projection specifically abuts on an inner face of the collar (19) of the ignition head (16).

2. The micro gas generator (10) according to claim 1, wherein the seat (24) for the igniter (14) is formed at least partly by a hollow-cylindrical portion (26) of the housing part (20).

3. The micro gas generator (10) according to claim 2, wherein a fixing cap (40) is attached to the hollow-cylindrical portion (26) in which the igniter (14) is received, or in that a fixing cap (40) is inserted in the hollow-cylindrical portion (26), the fixing cap retaining the igniter on the housing part.

4. The micro gas generator (10) according to claim 3, wherein an elevation (42), which biases the igniter (14) against the housing part (20), is formed on a bottom of the fixing cap (40).

5. The micro gas generator (10) according to claim 1, wherein a propelling charge container (46) is attached to the housing part (20), wherein the propelling charge container (46) is configured so that the igniter (14) is fixed in the seat (24) by the propelling charge container (46).

6. The micro gas generator (10) according to claim 1, wherein a stop is provided which delimits an insertion depth of the igniter (14).

7. The micro gas generator (10) according to claim 1, wherein a gasket (44) is interposed axially between the igniter (14) and the housing part (20).

8. The micro gas generator (10) according to claim 1, wherein the metallic insert (22) is a disk which projects radially from the housing part (20).

9. A method of manufacturing a micro gas generator (10) according to claim 1, comprising the steps of: placing the metallic insert in a tool mold, in the tool mold, the plastic housing part (20) is manufactured by injection-molding, wherein the metallic insert (22) is overmolded in such a way that it is positively connected to the housing part (20), and wherein the plastic housing part (20) includes the seat for the igniter, after injection-molding the igniter is inserted into the seat (24).

10. The micro gas generator (10) according to claim 4, wherein the elevation (42) is an annular elevation (42).

11. The micro gas generator (10) according to claim 7, wherein the gasket (44) is an O-ring.

12. A micro gas generator (10) for a vehicle safety system comprising a plastic housing part (20) and a metallic insert (22), wherein the plastic material forming the housing part (20) is injection-molded around the insert (22) so that the insert is positively connected to the housing part (20), and wherein a seat (24) in which an igniter (14) is inserted is formed in the housing part (20), the seat (24) for the igniter (14) being formed at least partly by a hollow-cylindrical portion (26) of the housing part (20), a fixing cap (40) being attached to the hollow-cylindrical portion (26) in which the igniter (14) is received, or being inserted in the hollow-cylindrical portion (26), wherein an elevation (42), which biases the igniter (14) against the housing part (20), is formed on a bottom of the fixing cap (40).

13. The micro gas generator (10) according to claim 12, wherein the elevation (42) is an annular elevation (42).

14. A micro gas generator (10) for a vehicle safety system comprising a plastic housing part (20) and a metallic insert (22), wherein the plastic material forming the housing part (20) is injection-molded around the insert (22) so that the insert is positively connected to the housing part (20), and wherein a seat (24) in which an igniter (14) is inserted is formed in the housing part (20), wherein a propelling charge container (46) is attached to the housing part (20), wherein the propelling charge container (46) is configured so that the igniter (14) is fixed in the seat (24) by the propelling charge container (46).

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